Prediction of Properties of Additively Manufactured Parts For Haynes 282

Undergraduate Project Report

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Abstract

Laser Powder Bed Fusion (LPBF) is one of the most advanced additive manufacturing techniques for fabricating high-performance metallic components with complex geometries. However, achieving defect-free parts with optimal mechanical properties requires precise control of process parameters such as laser power, scan speed, hatch spacing, and layer thickness.

This project presents a machine learning-based framework for predicting the properties of LPBF-fabricated Haynes 282 alloy components using experimentally collected and literature-derived datasets. A dataset consisting of approximately 500 samples was developed to establish relationships between process parameters and output properties including relative density, melt pool dimensions, hardness, porosity, and defect classification.

Multiple ensemble learning algorithms, including CatBoost Regressor, XGBoost Regressor, and LightGBM Regressor, were implemented and evaluated. Among the tested models, CatBoost demonstrated the best predictive performance with high accuracy and low error metrics across multiple properties.

Additionally, a process parameter optimization interface was developed to recommend suitable LPBF parameters for achieving desired target properties. The proposed framework provides a computationally efficient alternative to expensive experimental trials and numerical simulations, thereby contributing toward intelligent and data-driven additive manufacturing.

Chapter 1

Introduction

1.1 Background

Additive Manufacturing (AM), commonly referred to as 3D printing, has emerged as a transformative manufacturing technology capable of fabricating complex geometries directly from digital models. Unlike conventional subtractive manufacturing processes, additive manufacturing builds components layer-by-layer, thereby reducing material wastage and enabling unprecedented design flexibility.

Among the various additive manufacturing techniques, Laser Powder Bed Fusion (LPBF) has gained significant attention for the fabrication of high-performance metallic components used in aerospace, automotive, biomedical, and energy industries. In LPBF, a high-energy laser selectively melts metallic powder particles layer-by-layer according to the geometry defined by a CAD model.

The quality and performance of LPBF-manufactured components are highly dependent on process parameters such as laser power, scan speed, hatch spacing, and layer thickness. Improper combinations of these parameters can lead to defects such as porosity, lack of fusion, keyholing, and poor surface finish.

1.2 Haynes 282 Alloy

Haynes 282 is a nickel-based superalloy widely used in high-temperature applications due to its excellent creep resistance, oxidation resistance, thermal stability, and weldability. The alloy is particularly suitable for aerospace turbine components and high-temperature structural applications.

The alloy primarily consists of nickel, chromium, cobalt, molybdenum, titanium, and aluminum. The presence of titanium and aluminum promotes gamma-prime precipitation strengthening, while chromium enhances oxidation resistance.

Despite its excellent mechanical properties, processing Haynes 282 through LPBF

presents several challenges due to the complex thermal behavior involved during rapid melting and solidification.

1.3 Need for Process Optimization

The LPBF process involves a large parameter space, making experimental optimization both time-consuming and expensive. Traditional trial-and-error methods require extensive experimentation to determine suitable processing conditions for achieving high relative density and desirable mechanical properties.

Machine learning techniques offer an efficient alternative by learning relationships between input process parameters and output material properties from experimental and literature-derived datasets. Predictive models can significantly reduce experimental efforts and accelerate process optimization.

1.4 Problem Statement

In LPBF manufacturing, achieving consistent part quality remains a major challenge due to the nonlinear interaction between process parameters and material behavior. Small variations in laser power, scan speed, hatch spacing, or layer thickness can drastically influence melt pool characteristics, porosity formation, and mechanical properties.

Therefore, there is a need to develop a robust predictive framework capable of estimating material properties and recommending optimized process parameters for defect-free manufacturing.

1.5 Objectives

The primary objectives of this project are as follows:

1. To develop a dataset combining experimental and literature-derived LPBF process data for Haynes 282 alloy.

2. To predict properties such as relative density, melt pool width, melt pool depth, hardness, and defect class using machine learning techniques.

3. To compare the performance of different ensemble learning models including CatBoost, XGBoost, and LightGBM.

4. To generate printability maps for identifying stable and defective process regions.

5. To develop an inverse optimization framework capable of recommending suitable LPBF process parameters for desired output properties.

1.6 Scope of the Work

This work focuses on LPBF fabrication of Haynes 282 alloy using experimentally collected and literature-derived datasets. The study primarily investigates the influence of four process parameters:

- Laser Power
- Scan Speed
- Hatch Spacing
- Layer Thickness

The developed machine learning framework aims to provide accurate prediction and optimization capabilities while minimizing the need for extensive physical experimentation.

Chapter 2

Literature Review

2.1 Introduction

Laser Powder Bed Fusion (LPBF) has emerged as one of the most widely adopted metal additive manufacturing techniques due to its capability to fabricate highly complex geometries with excellent dimensional accuracy and mechanical performance. The process involves selective melting of metallic powder using a high-energy laser beam in a layer-by-layer manner.

In recent years, significant research efforts have focused on understanding the influence of LPBF process parameters on melt pool dynamics, porosity formation, microstructure evolution, and final mechanical properties. Machine learning and data-driven modeling techniques are also increasingly being explored for process optimization and property prediction.

This chapter presents a review of existing literature related to LPBF processing of Haynes 282 alloy, process parameter optimization, melt pool behavior, defect formation, and machine learning applications in additive manufacturing.

2.2 Laser Powder Bed Fusion Process

LPBF is a powder-bed additive manufacturing process in which thin layers of metallic powder are selectively melted using a focused laser beam. The process repeats layer-by-layer until the complete three-dimensional geometry is fabricated.

The major process parameters influencing the LPBF process include:

- Laser Power
- Scan Speed
- Hatch Spacing

- Layer Thickness

The interaction between these parameters governs the thermal history, melt pool geometry, cooling rate, and solidification behavior of the material.

The volumetric energy density is commonly used to characterize the combined effect of LPBF process parameters:

$$E = \frac{P}{vht} \tag{2.1}$$

where:

- E = Volumetric Energy Density
- P = Laser Power
- v = Scan Speed
- h = Hatch Spacing
- t = Layer Thickness

A suitable energy density range is critical for achieving dense and defect-free components.

2.3 Haynes 282 Alloy in Additive Manufacturing

Haynes 282 is a gamma-prime strengthened nickel-based superalloy developed for high-temperature structural applications, particularly in aerospace and gas turbine industries. The alloy exhibits excellent creep resistance, oxidation resistance, thermal stability, and weldability.

According to Haynes International, Haynes 282 possesses superior fabricability compared to conventional nickel-based superalloys while maintaining excellent high-temperature strength characteristics [1].

Shaikh et al. investigated the feasibility of manufacturing Haynes 282 using LPBF and demonstrated the successful fabrication of fully dense and crack-free components [2]. Their study reported that LPBF-fabricated Haynes 282 exhibited strong anisotropic grain structures due to directional solidification during layer-wise manufacturing.

The study further highlighted that the mechanical properties of LPBF-manufactured Haynes 282 at room temperature exceeded those of conventionally wrought material in certain conditions. However, anisotropy in tensile behavior and reduced ductility at elevated temperatures were observed.

Boswell et al. studied the influence of energy density and heat treatment on LPBF-manufactured Haynes 282 [3]. Their work demonstrated that heat treatment significantly reduced anisotropy by promoting recrystallization and equiaxed grain formation.

2.4 Defect Formation in LPBF

One of the major challenges in LPBF is the formation of defects caused by improper process parameter combinations. Common defects include:

- Lack of Fusion
- Keyholing
- Gas Porosity
- Balling
- Surface Irregularities

Lack of fusion defects occur when insufficient energy input prevents complete melting between adjacent tracks or layers. On the other hand, excessive energy input can lead to keyhole porosity caused by vaporization-induced cavity formation.

Khairallah et al. investigated pore formation mechanisms in LPBF and demonstrated the influence of recoil pressure, Marangoni convection, and melt pool instability on defect generation [4].

The study explained that stable melt pool formation is essential for minimizing porosity and ensuring uniform material consolidation.

2.5 Melt Pool Behavior

The melt pool is one of the most critical aspects of the LPBF process since it directly governs solidification behavior, microstructure evolution, and defect formation.

Melt pool dimensions such as width and depth are strongly influenced by laser power and scan speed. Higher laser power generally increases melt pool dimensions, whereas higher scan speed reduces interaction time and decreases melt pool size.

Several studies have shown that melt pool morphology can be used as an indicator of process stability and printability. Stable melt pools produce dense parts with reduced porosity, while unstable melt pools lead to irregular solidification and defect formation.

Recent advancements in in-situ monitoring and thermal imaging have enabled real-time observation of melt pool evolution during LPBF processing.

2.6 Machine Learning in Additive Manufacturing

Machine learning techniques are increasingly being adopted in additive manufacturing for process modeling, defect prediction, parameter optimization, and quality control.

Traditional finite element and physics-based simulations of LPBF are computationally expensive and time-consuming due to the complex thermal-fluid interactions involved during melting and solidification. Machine learning models provide a computationally efficient alternative by learning patterns directly from experimental and simulated datasets.

Various machine learning algorithms including Random Forest, Support Vector Machines, Artificial Neural Networks, Gradient Boosting, XGBoost, and CatBoost have been successfully applied for predicting material properties and process outcomes.

Liu et al. proposed a physics-informed machine learning framework for porosity prediction in LPBF [5]. Their study incorporated physical parameters such as energy density and recoil pressure into machine learning models to improve generalization capability.

Recent studies have also explored digital twin frameworks combining physics-based simulations and machine learning for real-time process optimization and defect monitoring.

2.7 Research Gap

Although significant research has been conducted on LPBF processing of nickel-based superalloys, several limitations still exist.

Most studies focus either on experimental characterization or isolated parameter optimization. Limited work has been reported on the development of unified datasets combining literature-derived and experimentally collected LPBF data for Haynes 282 alloy.

Furthermore, many studies primarily investigate individual output properties such as density or hardness, while comprehensive multi-property prediction frameworks remain limited.

There is also a need for inverse optimization systems capable of recommending process parameters for desired target properties. Such systems can significantly reduce experimental cost and accelerate industrial adoption of LPBF.

2.8 Summary

The literature review indicates that LPBF processing of Haynes 282 is a promising area of research due to the alloy's excellent high-temperature properties and printability characteristics. Previous studies have demonstrated the feasibility of producing dense and crack-free Haynes 282 components through careful process optimization.

Machine learning techniques have shown significant potential for process prediction and optimization in additive manufacturing. However, the development of generalized datasets, multi-property prediction frameworks, and inverse parameter recommendation systems remains an active research area.

The present work aims to address these gaps by developing a machine learning-based framework for prediction and optimization of LPBF process parameters and output properties for Haynes 282 alloy.

Chapter 3

Methodology

3.1 Overview

The methodology adopted in this work combines experimental data collection, literature-derived dataset generation, machine learning-based predictive modeling, and inverse process parameter optimization for Laser Powder Bed Fusion (LPBF) manufacturing of Haynes 282 alloy.

The overall workflow of the proposed framework is shown in Figure 3.1. The methodology begins with data acquisition from published literature and experimental measurements. The collected data is then preprocessed and used for training machine learning models capable of predicting multiple output properties from LPBF process parameters.

The developed framework also includes a printability classification approach and an inverse optimization system capable of recommending process parameters for desired target properties.

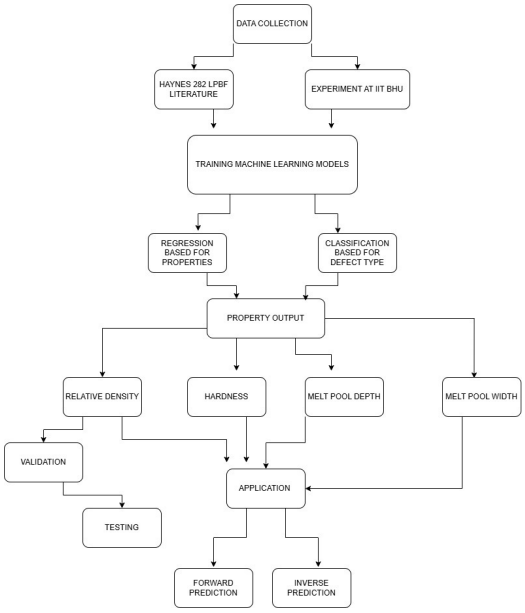


Figure 3.1: Overall methodology workflow adopted in this study

3.2 Data Collection

The dataset used in this study was developed using a combination of experimentally collected data and literature-derived process-property relationships.

The primary input process parameters considered in this work are:

- Laser Power (P)
- Scan Speed (v)
- Hatch Spacing (h)
- Layer Thickness (t)

The output properties predicted using machine learning models include:

- Relative Density
- Melt Pool Width
- Melt Pool Depth
- Hardness
- Porosity
- Defect Class

The experimental data was collected using LPBF-manufactured Haynes 282 samples fabricated on EOS M-series additive manufacturing systems. Additional data points were extracted from published research papers related to LPBF processing of Haynes 282 alloy.

3.3 Experimental Data Acquisition

Experimental samples were fabricated using varying combinations of LPBF process parameters in order to investigate the influence of processing conditions on final part quality.

The fabricated samples were characterized using density measurement techniques based on Archimedes' principle. Relative density values were obtained by comparing the sample weight in air and liquid medium.

The density was calculated using:

$$\text{Relative Density} = \frac{\left(\frac{A}{A-B}\right)}{\rho_{th}} \tag{3.1}$$

where:

- A = Weight of sample in air
- B = Weight of sample in liquid
- ρ_{th} = Theoretical density of Haynes 282 alloy

Thermal and optical observations during printing were also utilized for process monitoring and defect analysis.

3.4 Dataset Generation

A generalized dataset consisting of approximately 500 data points was developed by combining:

- Experimental measurements
- Literature-derived values
- Heuristically generated process-property relationships

The dataset contains the following features:

Table 3.1: Dataset Features Used in the Study

Feature	**Description**
P_W	Laser Power (W)
v_mm_per_s	Scan Speed (mm/s)
h_mm	Hatch Spacing (mm)
t_mm	Layer Thickness (mm)
MeltPoolWidth_um	Melt Pool Width (μm)
MeltPoolDepth_um	Melt Pool Depth (μm)
RelativeDensity_frac	Relative Density
Porosity_frac	Porosity Fraction
Hardness_HRA	Hardness (HRA)
DefectClass	Defect Classification

3.5 Volumetric Energy Density

The combined effect of LPBF process parameters was represented using volumetric energy density.

$$E = \frac{P}{vht} \tag{3.2}$$

where:

- E = Volumetric Energy Density
- P = Laser Power
- v = Scan Speed
- h = Hatch Spacing
- t = Layer Thickness

Volumetric energy density plays a critical role in determining melt pool stability and defect formation during LPBF processing.

3.6 Defect Classification

The dataset was classified into different printability regions based on process conditions and relative density behavior.

The following defect classes were considered:

- Stable
- Lack of Fusion
- Keyhole

Low energy density regions generally resulted in lack of fusion defects due to incomplete melting, whereas excessive energy density caused keyhole porosity because of vaporization-induced cavity formation.

The defect classification framework enabled the generation of LPBF printability maps for visualization of stable processing windows.

3.7 Data Preprocessing

Before model training, the dataset underwent preprocessing steps including:

- Missing value handling
- Feature normalization
- Outlier inspection
- Label encoding for defect classes
- Train-test data splitting

The dataset was divided into training and testing sets to evaluate model generalization performance.

3.8 Machine Learning Models

Three ensemble learning algorithms were implemented in this work:

- CatBoost Regressor
- XGBoost Regressor
- LightGBM Regressor

These algorithms were selected due to their strong nonlinear learning capability, robustness against overfitting, and excellent performance on tabular datasets.

3.8.1 CatBoost Regressor

CatBoost is a gradient boosting algorithm that efficiently handles categorical and numerical features while reducing prediction bias through ordered boosting techniques.

The algorithm demonstrated superior performance for predicting LPBF process-property relationships due to its ability to capture highly nonlinear interactions between process parameters.

3.8.2 XGBoost Regressor

XGBoost is a scalable gradient boosting framework widely used for regression and classification problems. It uses parallel tree boosting and regularization techniques to improve predictive accuracy and prevent overfitting.

3.8.3 LightGBM Regressor

LightGBM is a gradient boosting framework based on decision trees that utilizes histogram-based learning and leaf-wise tree growth strategies for improved computational efficiency.

3.9 Model Evaluation Metrics

The predictive performance of the developed machine learning models was evaluated using the following metrics:

3.9.1 Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \tag{3.3}$$

3.9.2 Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum |y_i - \hat{y}_i| \tag{3.4}$$

3.9.3 Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n}\sum (y_i - \hat{y}_i)^2} \tag{3.5}$$

3.9.4 Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{100}{n}\sum \left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{3.6}$$

3.10 Inverse Process Parameter Optimization

An inverse prediction framework was developed to recommend suitable LPBF process parameters for desired target properties.

The system accepts target outputs such as relative density, melt pool dimensions, and hardness as input and predicts corresponding process parameter combinations.

This approach significantly reduces experimental trials and enables rapid process optimization for industrial additive manufacturing applications.

3.11 Printability Map Generation

Printability maps were generated to visualize stable and defective process regions in the LPBF parameter space.

The maps were constructed using defect classification labels and volumetric energy density relationships. These maps help identify process windows that produce dense and defect-free components.

3.12 Summary

The proposed methodology integrates experimental characterization, dataset engineering, machine learning prediction, and inverse optimization into a unified framework for LPBF process analysis of Haynes 282 alloy.

The developed approach provides an efficient alternative to expensive trial-and-error experimentation and enables intelligent process optimization for additive manufacturing applications.

Chapter 4

Experimental Setup and Data Collection

4.1 Introduction

Experimental investigations were conducted to study the influence of Laser Powder Bed Fusion (LPBF) process parameters on the properties of Haynes 282 alloy components. The experimental setup involved fabrication of samples using an EOS metal additive manufacturing system followed by characterization and data acquisition for machine learning model development.

The collected experimental data was further combined with literature-derived data to develop a generalized process-property dataset for predictive modeling and optimization.

4.2 Material Description

Haynes 282 is a nickel-based superalloy designed for high-temperature structural applications due to its excellent creep resistance, oxidation resistance, and thermal stability.

The alloy primarily consists of nickel, chromium, cobalt, molybdenum, titanium, and aluminum. Titanium and aluminum contribute to gamma-prime precipitation strengthening, while chromium improves oxidation resistance at elevated temperatures.

The powder used for LPBF processing possessed a particle size distribution in the range of 20–55 μm, which is suitable for uniform powder spreading and stable melting behavior during LPBF fabrication.

4.3 LPBF System

The experimental samples were fabricated using EOS metal additive manufacturing systems available at the Precision Engineering Hub, IIT (BHU), Varanasi.

The EOS LPBF system operates by selectively melting metallic powder layers using a high-energy laser beam according to the sliced CAD geometry. Each deposited layer undergoes rapid melting and solidification, resulting in near fully dense metallic structures.

Figure 4.1 shows the EOS LPBF machine used during the experimental investigation.



Figure 4.1: EOS LPBF additive manufacturing system used for sample fabrication

4.4 Process Parameters

The primary LPBF process parameters investigated in this work are listed in Table 4.1.

Table 4.1: LPBF Process Parameter Range

Parameter	**Range**
Laser Power	150–370 W
Scan Speed	350–1772 mm/s
Hatch Spacing	0.07–0.12 mm
Layer Thickness	30–50 μm

Different combinations of these parameters were used to fabricate samples under varying thermal conditions for studying process-property relationships.

4.5 Sample Fabrication

Multiple test samples were fabricated using varying LPBF process parameters. The fabricated specimens included density cubes and process monitoring geometries designed to investigate printability and defect formation behavior.

Figure 4.2 shows the fabricated Haynes 282 samples after LPBF processing.

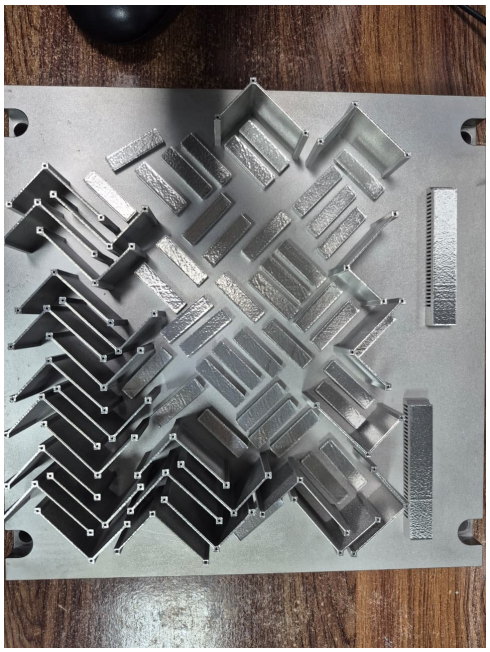


Figure 4.2: LPBF fabricated Haynes 282 samples

The fabricated samples exhibited varying surface morphologies and density characteristics depending on the selected process parameters.

4.6 Experimental Workflow

The complete experimental workflow involved:

1. CAD design preparation
2. Slicing and scan strategy generation
3. Powder deposition
4. Layer-wise laser melting
5. In-situ thermal and optical monitoring
6. Sample removal and cleaning
7. Density and property characterization

Figure ?? illustrates different stages of the experimental process.

4.7 Relative Density Measurement

Relative density measurements were performed using the Archimedes density measurement principle.

A Wensar analytical balance with high precision readability was used along with a density measurement kit. The density calculation was based on the difference between sample weight in air and weight in liquid medium.

The density equation is given as:

$$\text{Relative Density} = \frac{\left(\frac{A}{A-B}\right)}{\rho_{th}} \tag{4.1}$$

where:

- A = Weight of sample in air
- B = Weight of sample in liquid
- ρ_{th} = Theoretical density of Haynes 282 alloy

Figure 4.3 shows the density measurement setup used during experimentation.

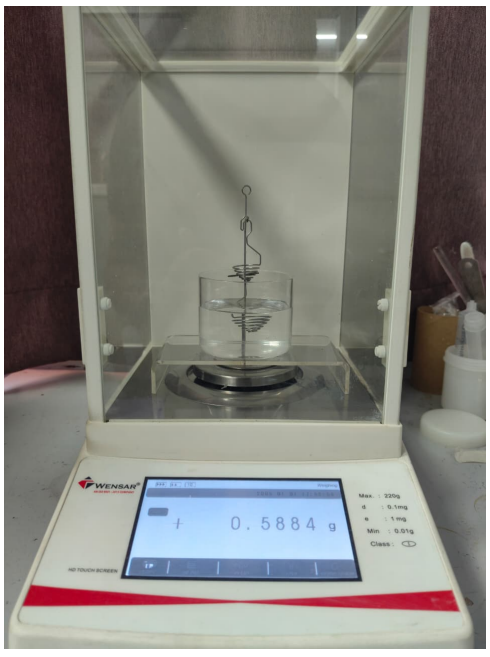


Figure 4.3: Density measurement setup based on Archimedes principle

4.8 Thermal and Optical Monitoring

Thermal and optical observations during LPBF processing were utilized to analyze process stability and melt pool behavior.

During fabrication, thermal signatures and optical images provided insight into:

- Melt pool formation
- Track continuity
- Spatter generation
- Defect formation
- Surface morphology evolution

The thermal observations also assisted in identifying stable and unstable processing conditions.

4.9 Dataset Compilation

The experimentally collected data was integrated with literature-derived datasets to generate a comprehensive process-property database for machine learning applications.

The final dataset consisted of approximately 500 samples and included:

- LPBF process parameters
- Melt pool dimensions
- Relative density
- Hardness
- Porosity fraction
- Defect classification

The generalized dataset improved model robustness and enabled multi-property prediction capabilities.

4.10 Printability Analysis

The experimentally observed process outcomes were categorized into stable and defective regions based on density measurements and melt pool behavior.

The following defect classes were identified:

- Stable Region
- Lack of Fusion
- Keyhole Region

Low energy density conditions generally produced lack of fusion defects due to incomplete melting, while excessive energy density led to keyhole instability caused by excessive vaporization and deep melt pool penetration.

The classified data was later used for generating LPBF printability maps.

4.11 Summary

This chapter presented the experimental setup and data collection methodology adopted for LPBF processing of Haynes 282 alloy.

The experimental investigations included sample fabrication, density measurement, thermal monitoring, and process characterization. The collected data formed the basis for the machine learning framework developed in this work for process prediction and optimization.

Chapter 5

Machine Learning Framework and Model Development

5.1 Introduction

Machine learning techniques were utilized in this work to establish relationships between Laser Powder Bed Fusion (LPBF) process parameters and output properties of Haynes 282 alloy. The developed framework enables prediction of material properties, defect behavior, and process optimization using experimentally collected and literature-derived datasets.

The machine learning pipeline involved data preprocessing, feature engineering, model training, hyperparameter tuning, performance evaluation, and inverse parameter optimization.

5.2 Machine Learning Workflow

The workflow consists of the following stages:

1. Dataset Collection
2. Data Preprocessing
3. Feature Engineering
4. Train-Test Split
5. Model Training
6. Hyperparameter Optimization
7. Model Evaluation
8. Prediction and Optimization

5.3 Dataset Description

The dataset developed for this work consisted of approximately 500 data points and 10 major features.

The input process parameters used for model training were:

- Laser Power (P)
- Scan Speed (v)
- Hatch Spacing (h)
- Layer Thickness (t)

The predicted output properties included:

- Relative Density
- Melt Pool Width
- Melt Pool Depth
- Hardness
- Porosity
- Defect Class

The dataset contained both experimentally collected and literature-derived data to improve generalization capability and prediction robustness.

5.4 Feature Engineering

Feature engineering was performed to improve the learning capability of the machine learning models.

One of the most important derived features used in this study was volumetric energy density.

$$E = \frac{P}{vht} \tag{5.1}$$

where:

- P = Laser Power
- v = Scan Speed

- h = Hatch Spacing
- t = Layer Thickness

Volumetric energy density captures the combined thermal effect of LPBF process parameters and plays a critical role in defect formation and melt pool behavior.

Additional preprocessing and feature engineering steps included:

- Label encoding of defect classes
- Numerical feature standardization
- Outlier inspection
- Missing value handling

5.5 Train-Test Splitting

The dataset was divided into training and testing subsets for evaluating model generalization performance.

Typically, 80% of the dataset was used for model training while 20% was reserved for testing and validation.

Cross-validation techniques were also utilized to improve reliability and reduce overfitting.

5.6 Machine Learning Algorithms

Three ensemble learning algorithms were implemented and compared in this work:

- CatBoost Regressor
- XGBoost Regressor
- LightGBM Regressor

These models were selected due to their strong capability in handling nonlinear regression problems and tabular datasets.

5.7 CatBoost Regressor

CatBoost is a gradient boosting algorithm developed specifically for handling categorical and numerical features efficiently while reducing prediction bias.

The algorithm utilizes ordered boosting techniques and symmetric decision trees to improve prediction stability and prevent overfitting.

The major advantages of CatBoost include:

- Strong nonlinear learning capability
- Reduced overfitting tendency
- Efficient handling of small datasets
- High predictive accuracy
- Robustness against noisy data

CatBoost demonstrated the best performance among the evaluated models for predicting LPBF process-property relationships.

5.8 XGBoost Regressor

XGBoost (Extreme Gradient Boosting) is a scalable ensemble learning framework based on gradient boosted decision trees.

The algorithm improves model performance through:

- Regularization
- Parallel tree boosting
- Gradient optimization
- Shrinkage techniques

XGBoost is widely used in regression and classification problems due to its computational efficiency and strong predictive capability.

5.9 LightGBM Regressor

LightGBM is a histogram-based gradient boosting framework designed for fast training and low memory usage.

Unlike traditional level-wise tree growth, LightGBM uses leaf-wise tree expansion, which improves computational speed and predictive performance.

The major advantages of LightGBM include:

- Fast training speed
- Lower memory consumption
- Efficient handling of large datasets
- High scalability

5.10 Hyperparameter Optimization

Hyperparameter tuning was performed to improve the predictive capability of the machine learning models.

The optimized parameters included:

- Learning Rate
- Number of Estimators
- Tree Depth
- Regularization Parameters
- Subsample Ratios

Grid search and cross-validation techniques were utilized for identifying optimal parameter combinations.

5.11 Model Evaluation Metrics

The predictive performance of the developed models was evaluated using multiple statistical metrics.

5.11.1 Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \tag{5.2}$$

The R^2 score measures the proportion of variance explained by the model. Higher values indicate better predictive accuracy.

5.11.2 Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum |y_i - \hat{y}_i| \tag{5.3}$$

MAE measures the average absolute prediction error.

5.11.3 Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2} \tag{5.4}$$

RMSE penalizes larger prediction errors and provides insight into prediction consistency.

5.11.4 Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{100}{n}\sum\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{5.5}$$

MAPE expresses prediction error as a percentage and is useful for comparing model performance across different properties.

5.12 Multi-Property Prediction

Separate machine learning models were trained for predicting:

- Relative Density
- Melt Pool Width
- Melt Pool Depth
- Hardness

The trained models demonstrated strong capability in learning complex nonlinear relationships between LPBF process parameters and output material properties.

5.13 Defect Classification

The developed framework also incorporated defect classification for identifying stable and defective process regions.

The defect classes considered were:

- Stable
- Lack of Fusion
- Keyhole

The classification labels were generated based on energy density behavior, porosity trends, and experimentally observed process conditions.

5.14 Inverse Process Optimization

An inverse optimization framework was developed to recommend LPBF process parameters for achieving desired target properties.

The framework accepts target outputs such as:

- Relative Density
- Melt Pool Width
- Melt Pool Depth
- Hardness

and predicts suitable combinations of:

- Laser Power
- Scan Speed
- Hatch Spacing
- Layer Thickness

This inverse approach significantly reduces experimental trials and enables rapid process optimization.

5.15 Summary

This chapter presented the machine learning framework developed for prediction and optimization of LPBF process parameters for Haynes 282 alloy.

The framework integrated data preprocessing, feature engineering, ensemble learning algorithms, model evaluation, and inverse optimization into a unified predictive system.

Among the implemented models, CatBoost demonstrated superior performance due to its strong nonlinear learning capability and robustness against overfitting.

Chapter 6

Results and Discussion

6.1 Introduction

This chapter presents the results obtained from the machine learning framework developed for prediction and optimization of Laser Powder Bed Fusion (LPBF) process parameters for Haynes 282 alloy.

The predictive capability of CatBoost, XGBoost, and LightGBM models was evaluated for multiple output properties including relative density, hardness, melt pool width, and melt pool depth.

The chapter also discusses printability behavior, defect classification, and inverse process parameter optimization.

6.2 Dataset Overview

The developed dataset consisted of approximately 500 data points containing LPBF process parameters and corresponding material properties.

The input features used for model training included:

- Laser Power
- Scan Speed
- Hatch Spacing
- Layer Thickness

The output properties predicted using machine learning models were:

- Relative Density
- Melt Pool Width

- Melt Pool Depth
- Hardness
- Porosity
- Defect Classification

The dataset combined experimentally collected values with literature-derived data in order to improve model generalization and robustness.

6.3 Relative Density Prediction

Relative density prediction is one of the most important objectives in LPBF process optimization since density directly influences the mechanical performance and structural integrity of manufactured components.

Table 6.1 shows the predictive performance of different machine learning models for relative density estimation.

Table 6.1: Model Performance for Relative Density Prediction

Metric	**CatBoost**	**XGBoost**	**LightGBM**
R^2	0.8713	0.7803	0.8177
RMSE	0.0022	0.0029	0.0026
MAE	0.0017	0.0023	0.0021
CV Score	0.8658	0.7617	0.8240
MAPE (%)	0.18	0.24	0.21

Among the evaluated models, CatBoost demonstrated the highest predictive accuracy with an R^2 score of 0.8713 and a low MAPE of 0.18%.

The results indicate that ensemble learning algorithms are highly effective in capturing nonlinear relationships between LPBF process parameters and density behavior.

6.4 Hardness Prediction

Hardness prediction was performed to estimate the mechanical behavior of LPBF-fabricated Haynes 282 components.

Table 6.2 summarizes the model performance for hardness prediction.

Table 6.2: Model Performance for Hardness Prediction

Metric	CatBoost	XGBoost	LightGBM
R^2	0.9686	0.7387	0.8130
RMSE	0.2770	0.7994	0.6763
MAE	0.1449	0.3468	0.3246
CV Score	0.8879	0.6872	0.7864
MAPE (%)	0.60	0.60	0.56

CatBoost significantly outperformed XGBoost and LightGBM for hardness prediction, achieving an R^2 score of 0.9686.

The results demonstrate the capability of gradient boosting algorithms in predicting complex material behavior influenced by LPBF thermal history.

6.5 Melt Pool Width Prediction

Melt pool geometry strongly influences microstructure evolution, porosity formation, and printability during LPBF processing.

Table 6.3 presents the predictive performance for melt pool width estimation.

Table 6.3: Model Performance for Melt Pool Width Prediction

Metric	CatBoost	XGBoost	LightGBM
R^2	0.9540	0.9291	0.9462
RMSE	5.7659	7.1608	6.2363
MAE	4.7378	5.8980	5.2210
CV Score	0.9522	0.9259	0.9473
MAPE (%)	3.33	4.16	3.61

The results indicate strong predictive capability of all three models, with CatBoost again achieving the best overall performance.

6.6 Melt Pool Depth Prediction

Prediction of melt pool depth is critical for understanding penetration behavior and layer bonding during LPBF processing.

Table 6.4 shows the predictive performance for melt pool depth.

Table 6.4: Model Performance for Melt Pool Depth Prediction

Metric	**CatBoost**	**XGBoost**	**LightGBM**
R^2	0.9726	0.9299	0.9640
RMSE	11.5158	18.4209	13.2021
MAE	8.8163	13.9329	9.0839
CV Score	0.9742	0.9476	0.9662
MAPE (%)	10.11	15.02	9.87

CatBoost demonstrated superior prediction capability for melt pool depth estimation with an R^2 value of 0.9726.

6.7 Experimental and Predicted Results

Table 6.5 presents a comparison between experimentally measured and machine learning predicted relative density values.

Table 6.5: Comparison of Experimental and Predicted Relative Density

Sample No.	Power (W)	Speed (mm/s)	Hatch (mm)	Thickness (mm)	Experimental Density	Predicted Density	Error (%)
1	260	1550	0.075	0.04	99.57	99.11	0.46
2	280	1450	0.075	0.04	99.53	99.30	0.23
3	270	1500	0.090	0.04	99.87	99.07	0.83
4	270	1500	0.080	0.04	99.11	99.07	0.04
5	270	1600	0.080	0.04	99.58	98.90	1.33

The low prediction error demonstrates the effectiveness of the developed machine learning framework.

6.8 Printability Map Analysis

Printability maps were generated using defect classification and process parameter relationships to identify stable and defective processing regions.

Figure 6.1 illustrates the printability behavior observed during LPBF processing of Haynes 282 alloy.

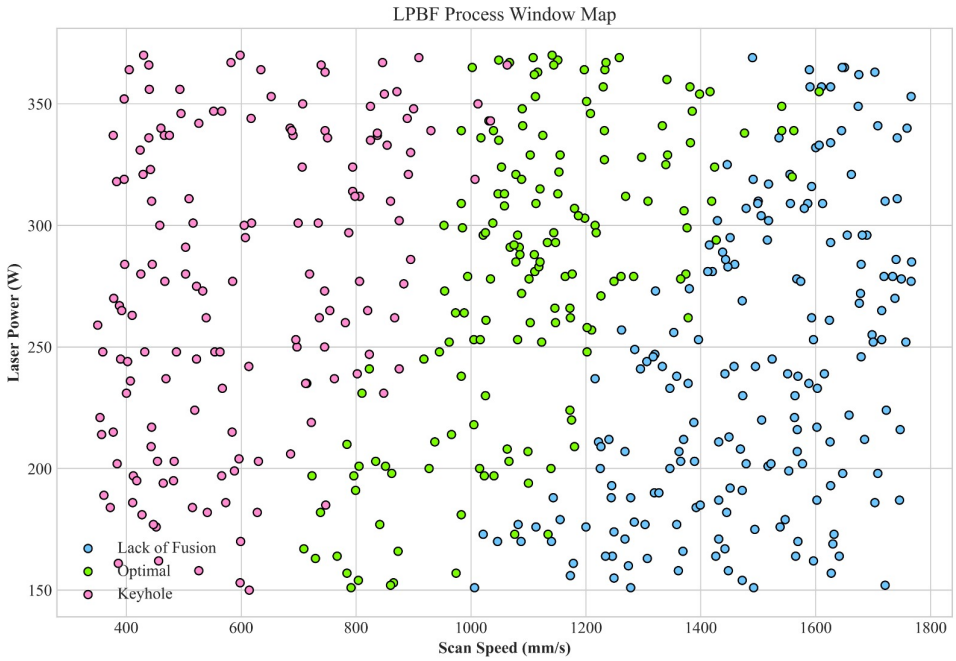


Figure 6.1: LPBF printability map for Haynes 282 alloy

The printability map clearly distinguishes:

- Stable process regions
- Lack of fusion regions
- Keyhole regions

Low energy density conditions produced insufficient melting and lack of fusion defects, whereas excessive energy density caused keyhole instability and deep vapor cavities.

6.9 Model Comparison

Among all evaluated machine learning algorithms, CatBoost consistently demonstrated superior performance across multiple prediction tasks.

The improved performance of CatBoost can be attributed to:

- Efficient nonlinear learning capability
- Ordered boosting strategy
- Reduced overfitting tendency
- Better handling of tabular datasets

XGBoost and LightGBM also showed strong predictive capability; however, CatBoost achieved the highest overall accuracy and generalization performance.

6.10 Inverse Process Parameter Optimization

An inverse optimization framework was developed to recommend LPBF process parameters for desired target properties.

The developed system enables users to input target properties such as:

- Relative Density
- Melt Pool Width
- Melt Pool Depth
- Hardness

and obtain optimized LPBF process parameter recommendations.

This approach significantly reduces experimental cost and enables rapid process optimization for industrial additive manufacturing applications.

6.11 Industrial Significance

The proposed framework has significant industrial applications in:

- Aerospace manufacturing
- Process optimization
- Intelligent additive manufacturing
- Real-time process monitoring
- Defect prediction
- Digital manufacturing systems

The developed machine learning models provide a computationally efficient alternative to expensive finite element simulations and extensive trial-and-error experimentation.

6.12 Summary

This chapter presented the results obtained from machine learning-based prediction and optimization of LPBF process parameters for Haynes 282 alloy.

Among the evaluated models, CatBoost demonstrated the best predictive performance across multiple material properties. The developed framework successfully predicted LPBF process outcomes and enabled inverse parameter optimization with high accuracy.

The generated printability maps and predictive models provide a valuable tool for intelligent additive manufacturing and process optimization.

Chapter 7

Conclusion and Future Scope

7.1 Conclusion

This work presented a machine learning-based framework for prediction and optimization of Laser Powder Bed Fusion (LPBF) process parameters for Haynes 282 alloy.

A generalized dataset consisting of approximately 500 data points was developed using a combination of experimentally collected measurements and literature-derived data. The dataset incorporated LPBF process parameters such as laser power, scan speed, hatch spacing, and layer thickness along with corresponding output properties including relative density, melt pool width, melt pool depth, hardness, porosity, and defect classification.

Multiple ensemble learning algorithms including CatBoost, XGBoost, and LightGBM were implemented and evaluated for predicting LPBF process-property relationships.

Among the evaluated models, CatBoost consistently demonstrated superior predictive capability across multiple output properties due to its strong nonlinear learning capability and robustness against overfitting.

The developed framework successfully achieved:

- Accurate prediction of relative density
- Prediction of melt pool dimensions
- Hardness prediction
- Defect classification
- Printability map generation
- Inverse process parameter optimization

The generated printability maps successfully identified stable and defective processing regions including lack of fusion and keyhole zones.

The inverse optimization framework further enabled prediction of suitable LPBF process parameters for desired target properties, thereby significantly reducing the need for expensive trial-and-error experimentation.

The developed machine learning system provides a computationally efficient alternative to traditional numerical simulations and has strong potential for intelligent additive manufacturing applications.

7.2 Major Contributions

The major contributions of this work are summarized as follows:

1. Development of a generalized LPBF process-property dataset for Haynes 282 alloy using experimental and literature-derived data.
2. Implementation and comparison of multiple ensemble learning algorithms for multi-property prediction.
3. Development of predictive models for relative density, melt pool dimensions, hardness, and defect behavior.
4. Generation of LPBF printability maps for identifying stable processing windows.
5. Development of an inverse optimization framework for recommending process parameters corresponding to desired target properties.
6. Integration of experimental observations with machine learning-based process optimization.

7.3 Industrial Applications

The proposed framework has several potential industrial applications in advanced manufacturing and aerospace industries.

Possible applications include:

- Intelligent process optimization
- Real-time quality prediction
- Defect minimization
- Digital manufacturing systems
- Aerospace component fabrication

- Data-driven additive manufacturing

The developed framework can significantly reduce material wastage, experimental cost, and optimization time in industrial LPBF systems.

7.4 Limitations of the Present Work

Although the developed framework demonstrated strong predictive capability, certain limitations remain.

- The dataset size is relatively limited compared to industrial-scale manufacturing databases.
- Some data points were generated using heuristic and literature-derived approximations.
- Thermal image analysis was limited to process monitoring and was not directly integrated into the machine learning pipeline.
- Residual stress and microstructural evolution were not explicitly modeled.
- Real-time closed-loop control was beyond the scope of the present work.

These limitations provide opportunities for future research and system improvement.

7.5 Future Scope

Several extensions and improvements can be implemented in future work.

7.5.1 Thermal Image-Based Deep Learning

Future work may involve integration of convolutional neural networks (CNNs) and deep learning models for direct analysis of thermal images captured during LPBF processing.

This can enable:

- Real-time defect detection
- Melt pool monitoring
- Dynamic process correction
- Automated quality control

7.5.2 Real-Time Closed Loop Control

Future research may focus on developing closed-loop control systems capable of dynamically modifying LPBF parameters during printing based on thermal feedback and defect prediction.

7.5.3 Physics-Informed Machine Learning

Combining physics-based thermal simulations with machine learning models can further improve prediction accuracy and generalization capability.

Physics-informed learning frameworks can incorporate:

- Heat transfer behavior
- Melt pool fluid dynamics
- Solidification physics
- Thermal gradients

7.5.4 Digital Twin Framework

The developed predictive framework can be extended toward digital twin systems for intelligent additive manufacturing.

A digital twin approach can enable:

- Real-time process monitoring
- Predictive maintenance
- Online optimization
- Manufacturing automation

7.5.5 Multi-Material Additive Manufacturing

Future work may also extend the developed methodology to:

- Multi-material LPBF systems
- Functionally graded materials
- Other nickel-based superalloys
- Titanium and aluminum alloys

7.6 Final Remarks

The integration of machine learning with additive manufacturing has significant potential for enabling intelligent and autonomous manufacturing systems.

The present work demonstrates that data-driven approaches can effectively predict LPBF process outcomes and optimize manufacturing parameters with high accuracy.

The developed framework provides a strong foundation for future research in intelligent additive manufacturing, real-time process optimization, and AI-assisted manufacturing systems.

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